

CO1101 Computing Fundamentals

Course Work Assessment-3

Submission deadline: Friday 15th December at 17:00

This is a group assessment. You are required to form groups of up to 6 students.

Assignment Tasks:

Write a short, structured report on one of the following topics:

- a. Home Automation through Internet of Things (IoT).
- b. Use of Data Analytics in Education.
- c. Use of Artificial Intelligence/Machine Learning in Healthcare Applications.

The report should contain a Title page, an Introduction/background section, a Literature review/Current state of the art section, a section on security and a conclusion. Please include your student numbers on the title page of submission.

There should be at least 4 good quality references used in the report. The references should be cited using IEEE referencing style.

Include high quality figures/diagrams in the write up.

The word count for the report is: ~1200 words $\pm 10\%$ (excluding references and title page)

1. **The Introduction/background:** this section should highlight the main idea behind technology and its applications. Some historical contexts should be provided here. This section should also provide a brief preface to what is contained in rest of the report.
2. **Literature Review:** Critical evaluation of the technology should be presented in this section which should also provide details of the current trends/directions the technology and its applications. Report on the technological advancements with respect to the specific topic. Details of tools and methods employed for applications in the area should also be highlighted here.
3. **Security, Ethical, and Social Issues:** This section should cover the security aspects of chosen topic. The security, ethical and social issues arising from use of the technology should be highlighted in this section.
4. **Conclusion:** This section should provide a summary of findings. What is the take home message of report? Provide reflection on the activity. How did you perform as a group? Are there any aspects of the assessment which you would like changed? What specific skills/knowledge is gained through completing the activity?

This submission forms a part of Leicester Award. In addition to the group report, **each student is required to submit two files (Transferrable Skills Audit and Personal Reflective Assessment form)**. There will be a separate submission link for these two files. These two files need to be put into a zip folder. Blank templates for these files are available to download from the assessment link.

Avoid plagiarism:

Any work you submit must be your own. If you use another's work and use it as your own, it is called plagiarism and it is a serious academic misconduct. You can use another person's work, but you must reference the material accurately. To avoid plagiarism, please follow a few rules:

- When you quote directly from another person's work, put the copied text inside quotation marks and **reference the source**.
- When you paraphrase or summaries another person's work, **reference the source**.
- When you refer to another person's ideas or theories, **reference the source(s)**, unless you are sure that they are common knowledge.
- When you copy a diagram, picture, graph, or table from someone else's work, **reference the source**.

Submission Instructions:

Submissions should be made through the module's blackboard page under the submission link.

Marking Criteria:

Criteria	Less than 40 (Fail)	40-49	50-59	60-69	Above 70
Introduction. (20%)	Weak introduction that fails to show the theme of the report.	Satisfactory basic introduction showing what the report is about.	Good introduction showing what the report is about	Very good introduction that clearly indicates what the report is about.	Excellent, very clear, concise, introduction covering all aspects of the report.
Effective written Communication (30%)	poor written style, difficult to understand. No reference.	Clear writing/formatting using average level of technical language with some references.	Effective writing/formatting using appropriate technical language with effective use of references	Excellent writing/formatting using excellent technical language and excellent use of relevant references.	Professional writing/formatting using excellent technical language and excellent use of relevant references.
Evidence of knowledge and understanding (30%)	Demonstrates little understanding of the topic/activity.	Satisfactory understanding of the topic with satisfactory grip of the topic/activity.	Good understanding of the topic with good grip of the topic/activity	Very good understanding of the topic with strong grip of the topic. Covers most of the aspects of the topic/activity.	Excellent and thorough understanding of the topic with comprehensive grip of the topic. Covers all aspects of the topic/activity with excellent critical analysis.
Tables, figures, and diagrams (10%)	Nothing presented	Lacking or are not sufficient with the text	Appropriate with the text but no references presented to support the finding	• Correct reference to figures from the text • Correct content and extent of legend • Results or numbers are correct and consistent	• Clear and complete displaying • The student made himself, non-existing, supportive figures
References (5%)	No references presented	Insufficient references, or to one-sided	References presented are satisfactory	• References are presented, sufficient, and correct.	• Important references are correctly integrated • Important references are correctly combined with own diagnostic findings, and this leads to depth.
Critical attitude (5%)	Is not questioning his/her own findings	Mixes up facts and opinions	Ask question due to own findings	Can separate facts and opinions	Critical review of the published work.